Predicting Primary School Completion Rate from GDP per Capita using Supervised Machine Learning

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1 Executive Summary

This project demonstrates the use of two supervised machine learning models, Linear Regression and Random Forest Regression, on a merged dataset containing the Primary School Completion Rates (PSCR) and GDP per capita of the different countries in line with the United Nations Sustainable Development Goal 4: Quality Education and the need for data-driven solutions. This project sourced its data from the World Bank and Our World in Data for Primary School Completion Rates and Our World in Data respectively then performed an inner merge and exploratory data analysis before creating the supervised machine learning models. Results from both models found that log(GDP per capita) is a predictor of Primary School Completion Rates (PSCR) and that the observed relationship reflects existing, long-term differences between high and low GDP countries. The researchers recommend using more robust models that account for the diminishing returns of increasing GDP.

2 Introduction

Sustainable Development Goal 4 (Quality Education) is a global priority because education is the foundation for improving lives and sustainable development (United Nations, 2015). The importance of this topic cannot be overstated, as education is a primary engine for economic growth. According to the Open Knowledge Repository (2018), higher levels of schooling are strongly linked to higher wages and increased national productivity. However, effectively solving these problems requires more than just good intentions. The United Nations Statistics Division (2023) stresses the critical need for data-driven solutions, arguing that without accurate data and statistical analysis, policymakers cannot see where progress is stalling or how to fix it.

To apply a data-driven approach to this issue, this project utilizes supervised machine learning, specifically a simple linear regression model. This project will analyze the relationship between Primary School Completion Rates (PSCR) and GDP per capita. By modeling the data, this project aims to move beyond theoretical assumptions and provide concrete statistical evidence of how primary education completion influences a nation's economic status.

This project attempts to answer the following research questions:

1. Is Primary School Completion Rate a predictor of GDP per capita?
2. What proportion of the variance of Primary Completion Rate is explained by GDP per capita? (The relationship between the two could be driven by either long term differences between different economic statuses of the respective countries or year-to-year changes in economic growth)

3 Exploratory Data Analysis

3.1 Data Sourcing and Methods

The data used in this project was sourced from The World Bank (for Primary School Completion Rates) and Our World in Data (for GDP per capita). The two datasets will be merged using an inner join on the common columns ”Country” and ”Year”. Furthermore, null and not-a-number(NaN) values are also dropped. A scatterplot will be created to visualize the distribution and to visually check for a linear relationship. Transformation will be performed when necessary before proceeding to creating the linear model.

3.2 Data Loading

Importing the libraries

```
import pandas as pd
import requests
from pandas_datareader import wb
```

Loading the World Bank Data for Primary School Completion Rates (PCSR)

```
df_pscr = wb.download(indicator='SE.PRM.CMPT.ZS',
          country='all', start=1970, end=2024)
df_pscr = df_pscr.reset_index()

# Rename columns for merging later
df_pscr = df_pscr.rename(columns={
    'country': 'Country',
    'year': 'Year',
    "SE.PRM.CMPT.ZS": 'PSCR'
})

# Year must be an integer
df_pscr['Year'] = df_pscr['Year'].astype(int)
```

Loading the data for GDP per capita from Our World in Data

```
df_gdp = pd.read_csv("https://ourworldindata.org/grapher/average
-years-of-schooling-vs-gdp-per-capita.csv?v=1&csvType=
full&useColumnShortNames=true", storage_options =
{'User-Agent': 'Our World In Data data fetch/1.0'})

#rename columns

df_gdp.rename(columns=
                {"Entity": "Country",
                "Year": "Year",
                "mys_sex_total": "Average Years of Schooling",
                "ny_gdp_pcap_pp_kd": "GDP per Capita"
                }, inplace=True)
```

3.3 Cleaning and Merging of Datasets

Merging the data on country and year using an inner join, then dropping null/NaN values for PSCR and GDP per Capita.

```
# Merge on Country and Year; exact match
if not df_pscr.empty and not df_gdp.empty:
    df_merged = pd.merge(df_pscr, df_gdp,
                         on=['Country', 'Year'],
                         how='inner')

# Drop rows with missing values in our target columns
(PSCR and GDP Per Capita)

df_clean = df_merged.dropna(subset=["PSCR", "GDP per Capita"])

df_clean.info()
```

Output:

```
    <class 'pandas.core.frame.DataFrame'>
Index: 3121 entries, 54 to 10229
Data columns (total 8 columns):
 #   Column                     Non-Null Count  Dtype
---  ------                     --------------  -----
 0   Country                    3121 non-null   object
 1   Year                       3121 non-null   int64
 2   PSCR                       3121 non-null   float64
 3   Code                       3121 non-null   object
 4   Average Years of Schooling 2905 non-null   float64
 5   GDP per Capita             3121 non-null   float64
 6   population_historical      3061 non-null   float64
 7   owid_region                110 non-null    object
dtypes: float64(4), int64(1), object(3)
memory usage: 219.4+ KB
```

3.4 Scatterplot for Distribution

```
import matplotlib.pyplot as plt
import seaborn as sns

#scatterplot of PSCR vs GDP per Capita
plt.figure(figsize=(10, 6))
sns.scatterplot(data=df_clean, x='PSCR', y='GDP per Capita')
plt.title('Primary School Completion Rate vs GDP per Capita')
plt.xlabel('Primary School Completion Rate (%)')
plt.ylabel('GDP per Capita (PPP)')
```

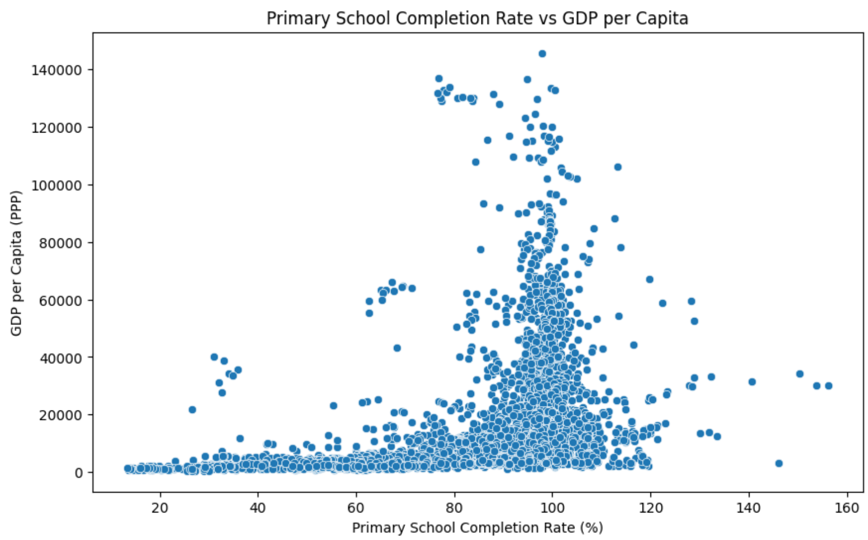

Figure 1: Initial Scatterplot of PSCR and GDP per Capita

Observe that Primary School Completion Rate (PSCR in %) values above 100. This is because PSCR in the context of this project is a gross ratio. According to the World Bank Indicator, the completion rate is defined as

$$Completion\ Rate = \frac{n_{graduating}}{N_i}$$

Where $n_{graduating}$ is the number of students graduating (any age) while N_i is the total population of the theoretical graduating age (i is the age). What this tells us is that if PSCR is greater than 100%, then it implies that the current education system is catching up on the backlog of students, while having PCSR less than 100% implies that children of the correct age appropriate to the graduating level are not graduating.

3.5 Transformation

A linear model has the assumption that it fits the equation of the line,

$$Y = \beta_0 + \beta_1 X + \epsilon$$

However, since the scatterplot shows a exponential curve as it approaches 100% rather than a straight line, the data will be transformed by taking the natural logarithm or ln(GDP per Capita). This is in line with standardized approaches for using GDP in relation to social outcomes and the law of diminishing returns (Preston, 1975; Stevenson & Wolfers, 2008).

```
import numpy as np
df_clean["log_GDP per Capita"] =
np.log(df_clean["GDP per Capita"])
```

Visualizing the transformed data,

```
plt.figure(figsize=(10, 6))
sns.scatterplot(data=df_clean, x='PSCR',
y='log_GDP per Capita')
plt.title('Primary School Completion Rate vs
Log of GDP per Capita')
plt.xlabel('Primary School Completion Rate (%)')
plt.ylabel('Log of GDP per Capita (PPP)')
plt.show()
```

Output

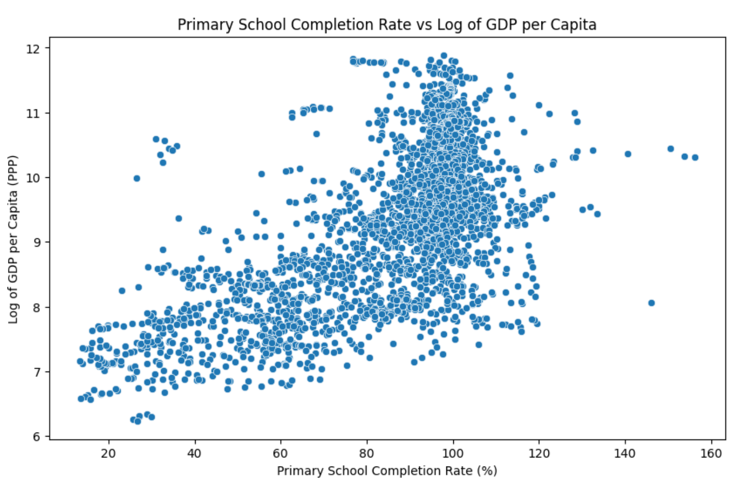

Figure 2: Transformed Data by taking the log(GDP)

4 Linear Model

4.1 Linear Regression

```
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score
X = df_clean[['log_GDP per Capita']] #predictor
y = df_clean['PSCR']

model = LinearRegression()
model.fit(X, y)
```

4.2 Predicting Test Data

```
y_pred = model.predict(X)
```

4.3 Results

```
#Solving for the equation of the linear model
slope = model.coef_[0]
intercept = model.intercept_
r2 = r2_score(y, y_pred)
```

Statistic	**Value**
Slope (β_1)	12.0290
Intercept (β_0)	-26.4807
R^2	0.4209

Table 1: Linear Regression Model Results

Interpretation: A 1% increase in GDP is associated with a 0.1203 point increase in completion rate. The R^2 value of 0.4209 implies a moderate relationship.

5 Machine Learning: Linear Regression

The previous chapter was a statistical analysis between GDP per capita and PSCR, which fits on 100% of the data. This chapter introduces supervised machine learning on predicting the education level of a country based on its GDP where the model is trained on 80% of the data, and tested on 20%.

5.1 Splitting the Data (80-20)

```
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error,
mean_absolute_error
```

Using the same parameters from the previous chapters,

```
X_train, X_test, y_train, y_test = train_test_split(
                                    X, y, test_size=0.2,
                                    random_state=42)
```

Creating a new linear model from the trained data,

```
lm = LinearRegression()
lm.fit(X_train, y_train)

y_pred_test = lm.predict(X_test)
```

Output

array([114.9801705 , 107.37746891, ... 102.78986279])

5.2 Results and Visualization

```
mse = mean_squared_error(y_test, y_pred_test)
root_mse = np.sqrt(mse)
r2_train = lm.score(X_train, y_train)
r2_test = r2_score(y_test, y_pred_test)
```

Statistic	**Value**
Model Intercept (β_0)	-26.58
Model Slope (β_1)	12.03
Training R^2 Score	0.4319
Testing R^2 Score	0.3802
MSE	322.7346
RMSE	17.96

Table 2: Machine Learning Model Results

Table 2 shows the results of the supervised machine learning evaluation. The Training R^2 score is how well the model learned the known data, while the Testing R^2 score corresponds to how well the model generalizes to new data. The RMSE or Root Mean Squared Error is the mean error in percentage. Given these scores, we can see that the model is not overfitting.

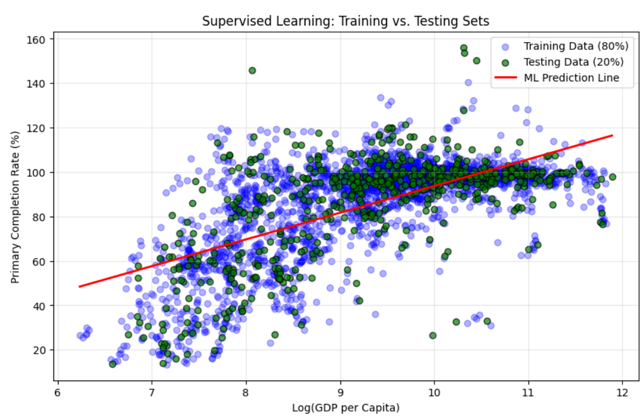

Figure 3: Training vs Testing Sets

Figure 3 shows the scatter plot and prediction line for the training data and the testing data.

Lastly, the residuals and its distribution are calculated for the model's validity.

$$Residuals = y - \hat{y}$$

```
resid = y_test - y_pred_test
plt.figure(figsize=(12, 5))
plt.subplot(1, 2, 1)
plt.scatter(y_pred_test, resid, alpha=0.6, color='purple')
plt.axhline(y=0, color='black', linestyle='--',
linewidth=2)
plt.title('Residuals vs. Predicted Values')
plt.xlabel('Predicted Completion Rate (PSCR) ')
plt.ylabel('Residuals (Error)')
```

Output

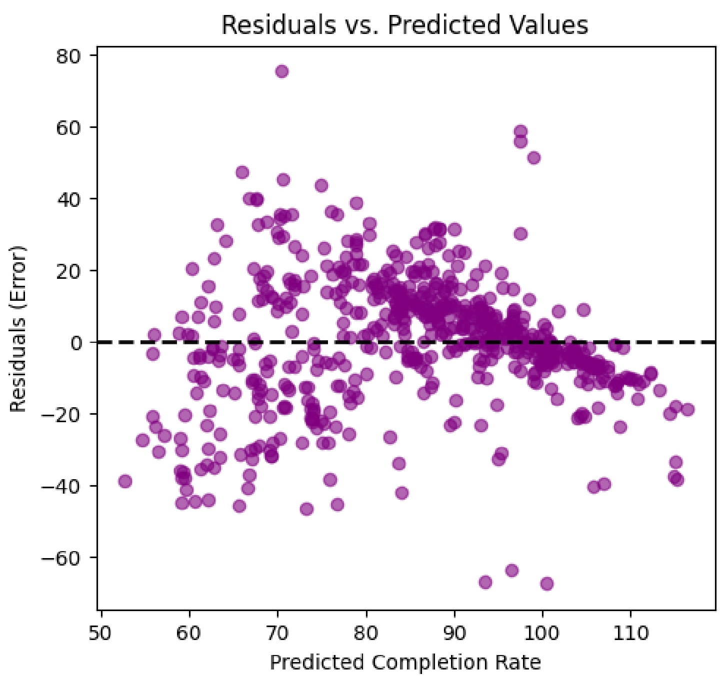

Figure 4: A Plot of the Residuals and Predicted Values

```
plt.subplot(1, 2, 2)
sns.histplot(resid, kde=True, color='teal', bins=20)
plt.title('Distribution of Residuals')
plt.xlabel('Residual Value')
plt.axvline(x=0, color='black', linestyle='--')
plt.tight_layout()
plt.show()
```

Output

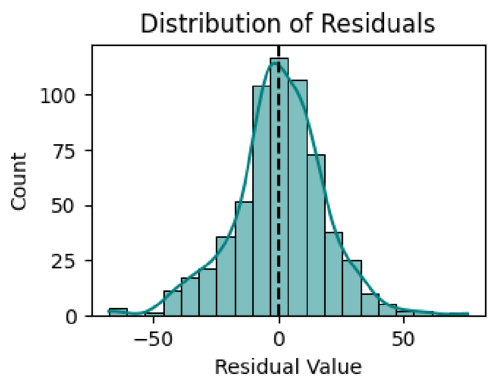

Figure 5: A KDE of the Distribution of Residuals

The residuals are (close to) normally distributed, centered at zero.

Statistic	**Value**
Mean of Residuals	0.4699
Std Dev. of Residuals	17.9587

Table 3: Mean and Standard Deviation of Residuals

6 Machine Learning: Random Forest Regressor

This project will also make use of a Random Forest Regressor to make use of a model that learns more complex patterns, being able to handle outliers better. The Random Forest Regressor creates multiple decision trees or mini-models and averages their predictions, allowing for the minimization of errors.

6.1 Splitting the Data

Using the same parameters as the previous chapter,

```
X = df_clean[['log_GDP per Capita']] #predictor
y = df_clean['PSCR']
X_train, X_test, y_train, y_test = train_test_split(
                                    X, y, test_size=0.2,
                                    random_state=42)
```

6.2 Model Training

Training the model with 100 decision trees,

```
rf_model = RandomForestRegressor(n_estimators=100,
                                random_state=42)
rf_model.fit(X_train, y_train)
```

6.3 Prediction and Evaluation

```
y_pred_rf = rf_model.predict(X_test)
rf_r2 = r2_score(y_test, y_pred_rf)
rf_rmse = np.sqrt(mean_squared_error(y_test, y_pred_rf))

print("Model Comparison")
print(f"Linear Model R^2:      {r2_test:.4f}")
print(f"Random Forest R^2:     {rf_r2:.4f}")
print(f"Random Forest RMSE:    {rf_rmse:.2f}")
```

Statistic	Value
Linear Model Testing R^2	0.3802
Random Forest R^2	0.3237
Training R^2 Score	0.4319
Random Forest MSE	352.1678
Random Forest RMSE	17.96

Table 4: Comparison between the Linear Model and Random Forest

6.4 Results and Visualization

```
plt.figure(figsize=(12, 6))

plt.scatter(df_clean['log_GDP per Capita'],
df_clean['PSCR'], color='blue', alpha=0.3,
label='Actual Data')
X_range = np.linspace(X['log_GDP per Capita'].min(),
X['log_GDP per Capita'].max(), 200).reshape(-1, 1)

#Linear model line
y_linear_range = lm.predict(X_range)
plt.plot(X_range, y_linear_range, color='red', linewidth=2,
linestyle='--', label='Linear Regression')
# Random Forest line
y_rf_range = rf_model.predict(X_range)
plt.plot(X_range, y_rf_range, color='green', linewidth=2,
label='Random Forest')

plt.title('Comparison: Linear Model vs. Random Forest')
plt.xlabel('Log(GDP per Capita)')
plt.ylabel('Completion Rate (%)')
plt.legend()
plt.grid(True, alpha=0.5)
plt.show()
```

Output

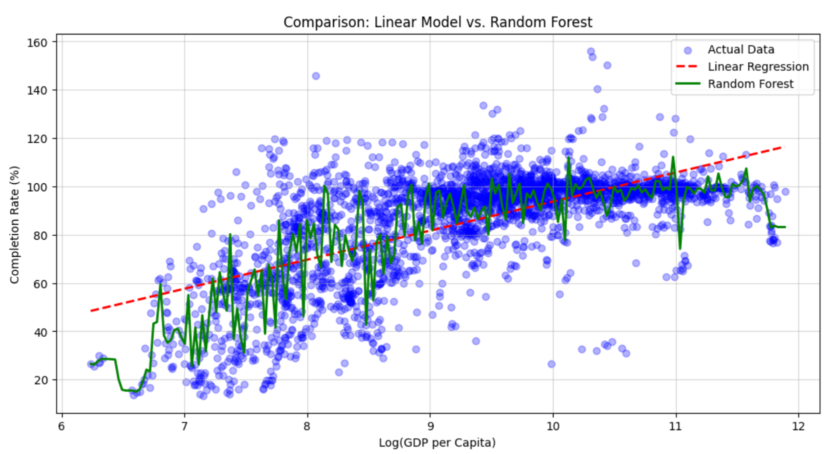

Figure 6: Comparison Between the Linear Model and Random Forest Regressor

From the figure, we can see the comparison between the Linear Model and the Random Forest Regressor. Observe that the Random Forest Line (the green line) follows the data better, even if the line itself becomes more jagged as it tries to follow the pattern of the data.

7 Conclusion

The results of both models show that there are clear diminishing results in increasing economic growth and PSCR as a metric of how educated a country's population is. Countries that are already rich see very little increase in PSCR, which was already hinted at by the initial scatterplot showing an increasing exponential curve. The linear model reported an R^2 value of 0.4209, implying that GDP per capita as the predictor only explained roughly 42% of the increase in PSCR. There are likely other factors or predictors at play that contribute to PSCR, though 42% for GDP per capita is already significant (since if we have multiple, other predictors, they have to share the remaining 58%). The observed relationship mostly reflects existing, long-term differences between high and low GDP countries rather than short term as expressed by year-to-year changes in economic growth.

The use of a linear model for GDP per capita and PSCR, which is rate, displays its flaws and limitations since as demonstrated in this paper, we end up having PSCR exceeding 100%. In the long run, completion rates cannot sustain a high percentage value, meaning that the models cannot predict a future with extreme economic growth.

8 References

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